

Muthuraja S

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CAREER OBJECTIVE

Passionate CS student who enjoys building things end to end. I have been exploring full-stack development through real projects, working mostly with the MERN stack and picking up tools like Redis, Docker, and CI/CD along the way. Getting into cloud deployment on GCP and AWS has been a big part of that too. I like working on problems where the code actually goes somewhere and does something.

EDUCATION

Indian Institute of Information Technology, Sri City, Tirupati, AP

Aug 2023 – May 2027

- B.Tech in Computer Science and Engineering

PROJECTS

CargoLink | [GitHub](#) | cargolink.muthuraja.tech

- Built a full-stack logistics marketplace using React, Vite, Express.js, and MongoDB on GCP App Engine; supports FTL/LTL booking, real-time shipment tracking via WebSockets, and Redis-based caching for frequently accessed order data.
- Set up authentication with JWT and OAuth 2.0, role-based access control, and IP rate limiting to protect the API; wrote integration tests using Jest on MongoDB Testcontainers.
- Set up GitHub Actions to run linting and tests on every pull request before deploying to App Engine; moving to Docker made the deployments consistent across environments.

Meta-Agent: LLM Orchestrator | [GitHub](#)

- Built a FastAPI engine that takes a natural language task description and figures out on its own which AI agents to run and in what order, instead of relying on manually written DAGs.
- Used NetworkX to build the execution graph and `asyncio.gather` to run independent agent groups at the same time, which cut end-to-end pipeline time noticeably.
- Plugged in Redis for state sharing between agents, ChromaDB for retrieval-augmented context, and OpenTelemetry to trace what each agent was doing during a run.

Parallel Vehicle & Traffic Analysis | [GitHub](#)

- Built a vehicle detection pipeline using OpenCV without any pretrained models – used frame differencing, CLAHE equalization, and morphological operations to spot vehicles across lane regions.
- Tuned detection separately for night, fog, and daytime conditions by adjusting V-channel thresholds and applying Dark Channel Prior for haze; got it working reliably across all three.
- Tracked vehicles using 8-connected blob analysis and centroid matching on a grid; used `multiprocessing.Pool` to handle frames in parallel and keep up with real-time video.

TECHNICAL SKILLS

Languages: Java, C, C++, JavaScript, Python, SQL, HTML/CSS

Frontend: React.js, Vite, Tailwind CSS, HTML5, CSS3

Backend: Node.js, Express.js, FastAPI, RESTful APIs

Databases & Caching: MongoDB, PostgreSQL, MySQL, Redis

DevOps & Testing: GitHub Actions, Docker, Jest, GCP App Engine, CI/CD

Core Competencies: OOP, Design Patterns, DBMS, Data Structures & Algorithms, API Design

POSITIONS OF RESPONSIBILITY

Web Development Domain Lead, Nexsync

08/2024 – Present

- Lead the web dev team for the college tech club; review PRs, make stack and architecture calls, and keep projects on track.
- Onboarded new members and ran internal sessions on Git workflows and REST API basics.

Design Team Core, Nexsync

09/2024 – 08/2025

- Designed posters, banners, graphics, and logos for organizational events.

Design Core, IIIT Sri City

09/2024 – 05/2025

- Created posters, banners, and social media graphics for college events.